

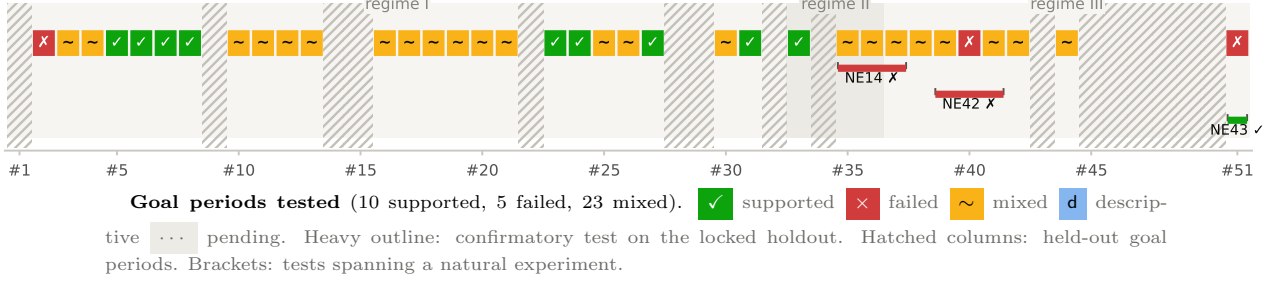
H19: One curve for all periods: per-period loop gains collapse onto a function of an operational control parameter

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1/5 Usefulness 2/5

Bottom line. Loop gains do not collapse onto x_{att} ; the round-1 “opposite channels” were the operator’s day edges, and trimmed gains move together.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

We pool per-period loop-gain estimates over 35 non-holdout periods: Hawkes and equal-time methods from H02–H05 and H19’s own. We ask whether they fall on one curve of x_{att} , the attended partners per unit attention. **The pre-registered collapse fails** in round 1 and again in round 1b. No common positive slope exists, and a regime-only step wins out of sample. Round 1 read two channels that move in opposite directions: activity co-activation rose with x_{att} while talk triggering fell. **Round 1b withdraws that reading** (corrected minute grid, day-edge-trimmed gains). The operator’s day edges made the activity rise: the slope goes $+0.18 \rightarrow -0.03$ after the trim. NE14 shows the same at the 2026-03-24 switch. After the trim, the activity gain agrees with the Hawkes gains. Post hoc, all six re-estimated methods rise with messages delivered per LLM step. The unfitted Hawkes-to-Ising mapping (P3) still holds.

PHYSICAL MODEL

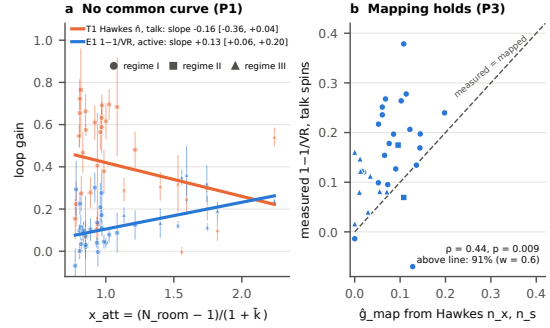
Mean-field Ising (01/02) and Hawkes (09). s_i : the 1-min active or talk spin of agent i . VR: variance ratio. g_{eq} : equal-time loop gain (H02’s βJ_{0q} , H04’s K , H05’s g). n_x , n_s : Hawkes cross- and self-offspring per event. w : the share within one $\Delta = 60$ s bin (kernel time τ_k). Dilution: per turn, $\bar{k}$ unread messages from $N_{room} - 1$ partners and a response budget b . y_{pm} : the gain of method m in period p . Control parameter: x_{att} .

$$g_{eq} = 1 - 1/VR, \quad x_{att} = (N_{room} - 1)/(1 + \bar{k})$$

$$VR = \text{Var}(\sum_i s_i) / \sum_i \text{Var} s_i, \quad n_x = b x_{att}$$

$$\hat{g}_{map} = \frac{2wn_x}{1 + 2w(n_x + n_s)}, \quad w = 1 - \frac{\tau_k}{\Delta} (1 - e^{-\Delta/\tau_k})$$

$$y_{pm} = \alpha_m + c_m x_{att,p} + u_p + \varepsilon_{pm}, \quad \text{collapse: } c_m > 0 \forall m$$



Observables. Round 1 (buggy minute grid, untrimmed). (a) The two primary loop gains per period vs x_{att} , with random-effects meta-regression lines. The Hawkes talk branching $\hat{n}$ (orange) falls and the equal-time active-spin gain (blue) rises. High x_{att} is mostly regime III (triangles). Round 1b shows that day edges make the blue rise (page 2). (b) The unfitted cross-method test: the equal-time talk gain vs the value mapped from H03’s fast Hawkes kernels ($w = 0.75$). Points sit above the identity line.

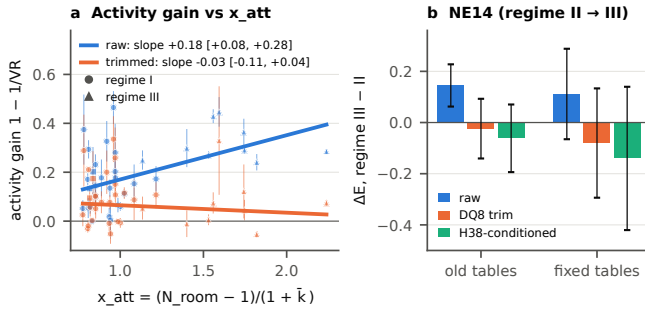
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Do not put one “loop gain” on a swarm dashboard: no single configuration knob predicts coupling strength. Trim the operator’s day edges before you compute any activity coupling. The edges carry about $\frac{3}{4}$ of the raw activity gain (NE43). After the trim, activity and talk gains move together. The one candidate knob is k_{llm} , the messages delivered per LLM step: all six methods rise with it. That knob is post hoc, entangled with the scaffold and fails inside #51, so it is a hypothesis to test, not a lever.

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RESULTS VS PREDICTIONS

Prediction	Observed (round 1 → 1b raw / 1b trimmed)	Verdict
P1 one curve on x_{att}	E1 slope $+0.13 \rightarrow +0.18/-0.03$; T1 -0.16 ; regime-only wins ($-11.5 \rightarrow -8.7/-7.3$ nats)	$\times$
P2 within-family $\rho \geq 0.4$	E: $-0.48 \rightarrow 0.21/0.28$ (0.44 H38- cond.); T: $0.40 \rightarrow 0.37$	$\times \rightarrow$ partial
P3 Hawkes→Ising map (unfitted)	ρ 0.44 $\rightarrow$ 0.37/0.33; above map 91% $\rightarrow$ 94/89%	$\checkmark$
P4 per-pair $n_x \propto (1 + \alpha_k \bar{k})^{-\alpha}$	0.20 $\rightarrow$ 0.26; on N : 1.03	$\times$
P5 no mode effect	CIs include 0	$\checkmark$
Regime III – I, activity gain	$+0.11 \rightarrow +0.17/-0.03$	gone
Native #51 N sweep	per-pair n_c falls far faster than $1/(N-1)$; trimmed g rises with N	$\times$
Native NE42 / NE43 / NE14	talk triggering ≈ 0 in the merged room; edges $\approx \frac{3}{4}$ of g on all NE43 sides; adjusted II→III < 0	$\times \checkmark \times$



Second figure. (a) Activity gain per period vs x_{att} , raw (blue) and day-edge-trimmed (orange), with meta-regression lines. The rise disappears after the trim. (b) NE14: the regime II→III change of the activity gain, raw vs trimmed vs H38-conditioned, on old and fixed tables.

GOAL PERIODS IN DETAIL

G02 $\times$ 1b: failed corrected data, pre-registered E1; with th...
G03 $\sim$ 1b: mixed corrected data, pre-registered E1; with the...
G04 $\sim$ 1b: mixed corrected data, pre-registered E1; with the...
G05 $\checkmark$ 1b: supported corrected data, pre-registered E1; with...
G06 $\checkmark$ 1b: supported corrected data, pre-registered E1; with...
G07 $\checkmark$ 1b: supported corrected data, pre-registered E1; with...
G08 $\checkmark$ 1b: supported corrected data, pre-registered E1; with...
G10 $\sim$ 1b: mixed corrected data, pre-registered E1; with the...
G11 $\sim$ 1b: mixed corrected data, pre-registered E1; with the...
G12 $\sim$ 1b: mixed corrected data, pre-registered E1; with the...

+28 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	2
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	0
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

H04's K and H05's gains used the buggy grid; we drop them until their owners re-run them, which leaves 6 methods. The trim keeps 77% of regime-III minutes, and the DQ8 trim and H38 conditioning disagree in places (NE42). The k_{llm} curve is post hoc and fails within #51. The methods share raw events, so the summed ELPD over-counts evidence. We did not re-run the synthetic validation of the P1 rule. The frozen confirmatory channel model uses round-1 data and is stale.

ROUND 2

Re-freeze the channel model from round 1b (talk and activity on k_{llm}) and pre-register it. Add H04/H05 after their re-runs. Test k_{llm} within periods where it varies at a fixed roster.

- **Essence.** The swarm has two order parameters, talk and work, that trade off.
- **H19-R1** Talk and work anticorrelate: periods and agents sit in talky or worky phases on a (talk, work) phase diagram.
- **H19-R2** Goal type (shared vs individual) moves the swarm between the two phases.
- **H19-R3** Messages per LLM step is the control parameter; pre-register the post-hoc collapse.

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collapse/README.md · ratings: RE-A2 re-evaluation agent (suggested), round 1b (page 2)